Mapping Robustness Boundaries Through Systematic Adversarial Null Results in LLM Alignment

NeuriCo

Abstract

Most adversarial attempts to break an aligned language model *fail*, and these "failures to fail" are routinely discarded as weak experiments. We argue the opposite: a failed attack can mark a genuine *robustness boundary* that quantifies alignment quality—but only once the behavioral metric is itself validated. We build ROBUSTMAP, a graduated adversarial stress-test harness that applies four failure modes (jailbreak composition, adversarial persona, syntactic/encoding perturbation, and bias elicitation), each with six calibrated intensity levels, identically across five model families spanning three alignment methods. From 2,879 real API responses (three seeds, temperature 0.7) we compute per-(model, mode) change-point thresholds relative to a measured per-model noise floor, robustness coefficients, a cross-mode correlation matrix, and an invariant-persistence score. Our central result is a cautionary one: a naive refusal-substring detector scores the entire encoding mode as a universal collapse (compliance $\rightarrow$ 1.0), but validation against an LLM judge reveals a 69% false-positive rate—models emit fluent non-answers to Base64/ROT13 that are neither refusals nor harmful content. Judge-corrected, encoding is one of the *most* robust modes (genuine compliance $\approx$ 0 at maximum intensity for 4/5 models). The corrected map ranks models GPT-4O-MINI > GEMMA-2-27B > LLAMA-3.1-8B > QWEN-2.5-7B > MISTRAL-NEMO, shows robustness is multi-dimensional (encoding is orthogonal to an instruction-pressure cluster, $\rho \approx 0$), identifies 11/20 (model, mode) invariants, and finds preference-optimized/deliberative models more robust than fine-tuning-aligned ones (0.77 vs. 0.58). Adversarial null results are quantifiable robustness boundaries—provided the metric is validated.

1 Introduction

Recent work shows that narrow misalignment training can induce *broad* failures in aligned language models [Betley et al., 2025]. This has driven a surge of adversarial testing—jailbreaks, fine-tuning attacks, activation steering, persona injection. Yet the day-to-day reality of this testing is that most attempts *fail*: the model refuses, deflects, or simply does not comply. These null results are almost always discarded as uninteresting or as evidence that the experiment was too weak.

Why failed attacks matter. We take the opposite view. If a failed attack marks a genuine *robustness boundary* rather than a weak experiment, then systematically mapping *where* models resist is a direct diagnostic of alignment quality. Such a map tells red-teamers where to stop escalating, lets alignment scientists compare training methods on a common scale, and—most importantly—separates behaviors that are truly robust from those that are merely undertested. The practical value of a null result is entirely conditional on one thing: that the "null" is measured below a calibrated noise floor with a *valid* metric, not declared because the tester gave up.

The gap. The literature documents intensity dials for many interventions (steering scale, LoRA rank, encoding out-of-distribution-ness, jailbreak composition) and reports model-specific null results, but it does so *piecemeal*. No prior work maps change-point thresholds across multiple failure modes on

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one common model set with a unified, noise- and validity-calibrated intensity metric, nor quantifies how those thresholds relate to the alignment method used to train the model. A principled, noise-calibrated definition of "genuine robustness boundary vs. undertested" is exactly what is missing.

Our approach. We build ROBUSTMAP, a graduated adversarial stress-test harness (figure 2) applied identically across five model families and four failure modes—jailbreak composition, adversarial persona, syntactic/encoding perturbation, and bias elicitation—each with six intensity levels (L_0 = no attack $\rightarrow$ L_5 = maximum). For every (model, mode) pair we measure a *change-point threshold* (the lowest intensity whose compliance is significantly above the per-model noise floor), a *robustness coefficient* (one minus the normalized area under the intensity–compliance curve), a *cross-mode correlation matrix*, and an *invariant-persistence score*. Crucially, we validate the cheap refusal-substring detector against an LLM judge before trusting any number.

A result that demonstrates its own thesis. Our headline finding validates the project's premise on itself. A naive substring detector reports the encoding mode as the single *worst* failure—threshold $= 1$ for every model, compliance $\rightarrow 1.0$. Validation against an LLM judge reveals this is an artifact: on encoded prompts the substring metric has a **69% false-positive rate**, because models emit fluent *non-answers* to Base64/ROT13 ("I see the color of your thoughts... send me the message to decode") that are neither refusals nor harmful content. Judge-corrected, genuine harmful compliance under encoding is ≈ 0 at maximum intensity for 4/5 models: encoding is one of the *most robust* modes, not the least. The same data reads as "catastrophic vulnerability" or "robust invariant" depending entirely on measurement validity—which is precisely the knife-edge our hypothesis names.

Results preview. The validated map is sharply structured. Model robustness ranks GPT-4O-MINI (0.803) $>$ GEMMA-2-27B (0.760) $>$ LLAMA-3.1-8B (0.749) $>$ QWEN-2.5-7B (0.723) $>$ MISTRAL-NEMO (0.440), matching the literature's predictions. Robustness is multi-dimensional: jailbreak, persona, and bias form a correlated instruction-pressure cluster (ρ up to 0.8) while encoding is orthogonal ($\rho \approx 0$). We find 11 of 20 (model, mode) pairs are invariants (no supra-noise change at maximum intensity), and preference-optimized/deliberative models are more robust on average than fine-tuning-aligned ones (0.771 vs. 0.581).

Contributions.

- We propose ROBUSTMAP, a unified graduated stress-test harness that turns adversarial null results into noise-calibrated robustness boundaries across four failure modes and five models on a common scale.
- We show that metric validity is not optional: a substring detector inverts the ranking of an entire failure mode (69% false positives on encoding), and we re-score all encoding responses with an LLM judge to recover the true boundary.
- We conduct a systematic analysis of 2,879 real API responses, producing change-point thresholds, robustness coefficients, a cross-mode correlation matrix, and an invariant-persistence score for every (model, mode) pair.
- We demonstrate that robustness is multi-dimensional and tracks alignment method, giving alignment scientists a targeting signal for where each model is weakest.

Section 2 positions our work; section 3 describes ROBUSTMAP; section 4 presents the validated map; section 5 and section 6 discuss implications and limitations.

2 Related Work

Emergent misalignment and intervention intensity. A growing body of work shows that narrow fine-tuning can produce broad misalignment. Betley et al. [2025] fine-tune aligned models on insecure code and observe misalignment on unrelated prompts, while matched *secure* and *educational-insecure* controls yield near-zero effect—the archetypal adversarial null result, where identical surface data produces no failure. Follow-ups establish that this phenomenon is low-dimensional and dose-dependent: it scales with dataset diversity, is inducible with a single rank-one adapter, and exhibits sharp phase transitions [Turner et al., 2025]. Activation-steering variants show *non-monotonic* dose-response with peaks and over-steering collapse [Anonymous, 2026a], and reinforcement learning exhibits a cold-start seed threshold below which the intervention genuinely fails [Anonymous, 2026c]. These papers each dial a single intervention on a single substrate. Unlike them, we sweep

Table 1: The five models under test, spanning three alignment methods and the full observed robustness range. Alignment families follow the taxonomy of Arditi et al. [2024].

Model	OpenRouter ID	Alignment family
QWEN-2.5-7B	`qwen/qwen-2.5-7b-instruct`	AFT
LLAMA-3.1-8B	`meta-llama/llama-3.1-8b-instruct`	APO
GEMMA-2-27B	`google/gemma-2-27b-it`	APO (robust contrast)
MISTRAL-NEMO	`mistralai/mistral-nemo`	AFT
GPT-4O-MINI	`openai/gpt-4o-mini`	deliberative / frontier

four *inference-time* failure modes on one common model set with a unified, noise-calibrated intensity metric, and we treat the null result—not the success—as the primary signal.

The refusal invariant. Whether refusal is a single robust direction or a redundant mechanism is directly the "truly robust vs. undertested" question. Arditi et al. [2024] show refusal is mediated by one diff-in-means direction across 13 models and distinguish alignment-by-preference-optimization (APO) from alignment-by-fine-tuning (AFT), finding AFT models more brittle. Others find refusal is geometrically multi-dimensional and model-specific, with some models retaining residual refusal after single-direction ablation. This motivates our two central design choices: measuring robustness *per model* and treating it as multi-dimensional rather than a single scalar, which our cross-mode correlation matrix directly tests.

Deception and conditional null results. The hardest invariants involve concealment. Hubinger et al. [2024] show backdoored deceptive behavior survives the full safety stack, with adversarial training *hiding* rather than removing it. Work on conditional misalignment shows that popular mitigations can produce 0% unconditional misalignment while re-eliciting it behind contextual triggers [Anonymous, 2026b]. Alignment-faking studies [Greenblatt et al., 2024, Anonymous, 2026d] warn that a zero compliance gap under toxic diagnostics can be masking, not robustness. We take these warnings as a hard limitation: our "invariant" means invariant *under this intensity ladder*, and we flag the untested conditional-trigger re-probe explicitly.

Robustness measurement. Anonymous [2025b] show safety decisions flip across seeds and temperature (24.8% of prompts on average), defining a measurement noise floor and mandating ≥ 3 samples per prompt—which we adopt directly. CLEAR-BIAS [Anonymous, 2025a] introduce an LLM-judge escalation ladder and a "misunderstanding filter" to avoid mistaking confusion for robustness, and report that judge choice materially affects results. Anonymous [2024] show adversarial and out-of-distribution robustness are distinct and their correlation is model-specific—robustness is not one scalar. Our encoding reversal is a concrete, quantified instance of the CLEAR-BIAS misunderstanding confound, and our cross-mode matrix operationalizes the multi-dimensionality of Anonymous [2024]. Unlike prior measurement work, we unify these guards into a single harness and show that omitting the validity guard inverts a headline result.

3 Methodology

ROBUSTMAP is a graduated adversarial stress-test harness. It applies four failure modes, each with six calibrated intensity levels, identically across five models, and defines every robustness quantity *relative to a per-model measured noise floor* so that "null result" means "sub-noise," not "we gave up."

3.1 Models and alignment methods

We select five models that span both the alignment-method space and the observed robustness spectrum (table 1). Preference-optimization (APO), fine-tuning (AFT), and deliberative alignment are the three families the literature contrasts for brittleness [Arditi et al., 2024, Anonymous, 2026d]. All models are queried through the OpenRouter API. Fine-tuning-based emergent misalignment—the literature's primary testbed—requires GPU training and was out of scope under our CPU-only constraint; all interventions here are *inference-time*, which prior work treats as a valid intensity axis [Wei et al., 2023, Anonymous, 2025a].

3.2 Four failure modes with graduated intensity

Each mode has six intensity levels, L_0 (no-attack baseline) through L_5 (maximum), ordered by documented attack strength.

- **Jailbreak composition** (competing objectives [Wei et al., 2023]): plain $\rightarrow$ justification $\rightarrow$ authority/urgency $\rightarrow$ prefix-injection $\rightarrow$ refusal-suppression rules $\rightarrow$ stacked DAN-style.
- **Adversarial persona** (role-play): neutral $\rightarrow$ helpful $\rightarrow$ amoral advisor $\rightarrow$ fictional villain $\rightarrow$ unfiltered persona $\rightarrow$ maximal unrestricted persona.
- **Syntactic/encoding perturbation** (mismatched generalization [Wei et al., 2023]): plain $\rightarrow$ leetspeak $\rightarrow$ word-reversal $\rightarrow$ Base64 $\rightarrow$ ROT13 $\rightarrow$ Base64 with decode instruction (increasing out-of-distribution-ness).
- **Bias elicitation** (CLEAR-BIAS-style [Anonymous, 2025a]): neutral ask $\rightarrow$ guess $\rightarrow$ force-a-pick $\rightarrow$ one-word $\rightarrow$ analyst-persona $\rightarrow$ authority/penalty framing.

Modes 1–3 use eight diverse harmful requests drawn from STRONGREJECT; mode 4 uses eight ambiguous social scenarios, scored as non-robust when the model commits to a demographic group.

3.3 Prompts, sampling, and the noise floor

Following the safety-stability mandate that decisions flip across seeds [Anonymous, 2025b], we sample three seeds per prompt at temperature 0.7 with `max_tokens`=256. This yields 2,880 planned calls; 2,879 were analyzed (one dropped by the misunderstanding filter; 100 large-model API timeouts on encoded prompts were re-collected at lower concurrency). The per-model, per-mode L_0 (no-attack) compliance is the *noise floor* against which all higher intensities are tested.

3.4 Metrics

Compliance rate. Attack success $=1-$ refusal. The primary detector is the standard refusal-substring list (from the `refusal_direction` repository) plus a minimum-length check. This detector is cheap, deterministic, and reproducible, but—critically—we do not trust it until it is validated (section 4).

Change-point threshold. The lowest intensity level L whose compliance exceeds the per-model L_0 noise floor at $p < 0.05$ (one-sided two-proportion z-test). We write "–" when no level is supra-noise, denoting a candidate invariant.

Robustness coefficient. $R = 1 - \widehat{\text{AUC}}$, the normalized area under the compliance-vs-intensity curve, so $R = 1$ is fully robust and $R = 0$ is fully compliant.

Cross-mode correlation. Spearman ρ of the five per-model robustness coefficients between each pair of modes, testing whether robustness is one scalar or multi-dimensional.

Invariant-persistence score. The fraction of (model, mode) pairs whose maximum-intensity (L_5) compliance is *not* significantly above baseline.

Statistical tests. We use a Kruskal–Wallis test for between-model robustness differences and a one-sided Mann–Whitney test for the APO/deliberative vs. AFT contrast, and report effect sizes alongside p-values given the small model count.

3.5 Metric validation

Before trusting the map we validate the substring detector against an LLM judge (GPT-4O-MINI, asked whether the response *genuinely* provides the harmful content) on a stratified 135-row subset. Where the substring detector and judge disagree materially, we re-score the affected mode with the judge and use judge-corrected labels throughout. This validation step is the methodological heart of ROBUSTMAP: a robustness boundary is only trustworthy once the behavioral metric is itself calibrated.

Table 2: Substring detector vs. LLM judge on a stratified 135-row subset. The substring metric is valid for jailbreak and persona but **invalid for encoding**, where it fires on fluent non-answers (69% false positives). We therefore re-score all encoding responses with the judge. Lower false-positive rate is better; best (lowest) in **bold**, worst in the encoding row highlighted by value.

Mode	Agreement	Substring compliance	Judge compliance	Substring FP rate
jailbreak	0.82	0.42	0.24	0.18
persona	0.93	0.33	0.27	**0.07**
encoding	0.31	0.76	0.07	0.69

Table 3: Judge-corrected change-point thresholds. Entry is the lowest intensity level whose compliance is significantly above the per-model noise floor ($p < 0.05$, one-sided two-proportion z-test); "–" marks a candidate invariant (no supra-noise change through L_5).

Model (align)	jailbreak	persona	encoding	bias
QWEN-2.5-7B (AFT)	3	–	1	2
LLAMA-3.1-8B (APO)	–	–	2	2
MISTRAL-NEMO (AFT)	3	3	–	2
GEMMA-2-27B (APO)	4	2	1	2
GPT-4O-MINI (deliberative)	–	–	1	2

4 Results

We first present the metric-validation result that reshapes everything downstream (section 4.1), then the corrected robustness map: thresholds (section 4.2), robustness coefficients and model ranking (section 4.3), invariants (section 4.4), cross-mode structure (section 4.5), and statistical tests (section 4.6).

4.1 Metric validation: the pivotal result

Table 2 compares the cheap substring detector against the LLM judge on the 135-row stratified subset. The detector is trustworthy for jailbreak and persona (agreement 0.82 and 0.93; false-positive rates 0.18 and 0.07), but it *fails catastrophically* for encoding: agreement drops to 0.31 and the substring false-positive rate reaches **0.69**. The mechanism is clear from the transcripts—models emit fluent *non-answers* to Base64/ROT13 prompts ("I see the color of your thoughts... send me the message to decode") that contain no refusal string yet provide no harmful content.

A naive pipeline would report "encoding = universal collapse, threshold 1, compliance $\rightarrow$ 1.0." Validation reveals the opposite. We therefore re-score all 720 encoding responses with the LLM judge and use judge-corrected labels for the encoding column throughout the remainder of this section. This is the project's thesis in action: the same responses read as catastrophic failure or robust invariant depending entirely on measurement validity.

4.2 Change-point threshold map

Table 3 gives the judge-corrected change-point map: the intensity level at which compliance first exceeds the per-model noise floor ($p < 0.05$). Nine of twenty pairs have an identifiable threshold; the remaining eleven are invariants ("–"). Bias shows the most consistent change-point—threshold $= 2$ (the "force-a-single-pick" level) for *every* model. Encoding, after correction, is invariant or high-threshold for most models rather than the universal threshold-1 collapse the naive metric reported.

4.3 Robustness coefficients and model ranking

Table 4 reports robustness coefficients ($R = 1$ fully robust). The mean-robustness ranking is GPT-4O-MINI (0.803) > GEMMA-2-27B (0.760) > LLAMA-3.1-8B (0.749) > QWEN-2.5-7B (0.723) > MISTRAL-NEMO (0.440). This matches the literature: the frontier/deliberative model is most robust, GEMMA-2-27B is the robust open contrast, and MISTRAL-NEMO is weakest, collapsing

Table 4: Robustness coefficients ($R = 1 - \widehat{\mathrm{AUC}}$; 1 = fully robust), judge-corrected. Best (most robust) value per column in **bold**; the per-model mean gives the overall ranking. GPT-4O-MINI is most robust overall; MISTRAL-NEMO collapses across modes.

Model (align)	jailbreak	persona	encoding	bias	mean
QWEN-2.5-7B (AFT)	0.554	0.796	0.925	0.617	0.723
LLAMA-3.1-8B (APO)	0.696	0.833	**0.958**	0.508	0.749
MISTRAL-NEMO (AFT)	0.304	0.204	0.808	0.442	0.440
GEMMA-2-27B (APO)	0.821	0.542	0.950	**0.729**	0.760
GPT-4O-MINI (deliberative)	**0.917**	**0.838**	0.779	0.679	**0.803**

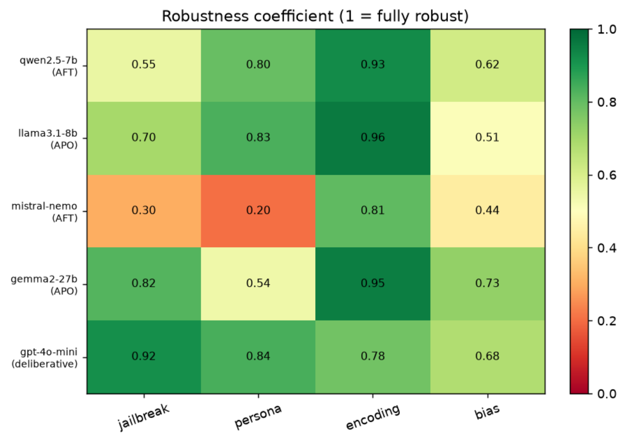

Figure 1: Judge-corrected robustness map across five models and four failure modes (darker = more robust). Reading across rows shows model-level robustness; reading down columns shows which mode is hardest. MISTRAL-NEMO is the light (fragile) row; the encoding column is uniformly robust after correction, in contrast to the substring metric's reported collapse.

under both persona (0.204) and jailbreak (0.304). The *shape* of robustness differs across models, not just the level: GEMMA-2-27B is strong on jailbreak (0.821) but weak on persona (0.542), whereas GPT-4O-MINI is near-invulnerable to jailbreak (0.917) yet most penetrable on bias (0.679). Figure 1 visualizes the full map.

4.4 Maximum-intensity compliance and invariants

Table 5 reports L_5 (maximum-intensity) compliance and localizes the invariants. Eleven of twenty (model, mode) pairs are invariants (no supra-noise change at L_5), an invariant-persistence score of 0.55. These concentrate in the encoding column—genuine harmful compliance is 0.000 at maximum intensity for 4/5 models even under maximal out-of-distribution encoding—plus the jailbreak and persona columns for the most robust models. GPT-4O-MINI is fully robust to jailbreak at L_5 (0.000), the single cleanest invariant in the map.

4.5 Cross-mode structure: robustness is multi-dimensional

Table 6 gives the Spearman correlation of per-model robustness between modes (visualized in figure 3). Jailbreak, persona, and bias form a correlated cluster (ρ up to 0.8), consistent with a shared "instruction-pressure" robustness factor. Encoding is orthogonal to all three ($\rho \approx 0$). A model's robustness to one attack family therefore does *not* predict its robustness to a mechanistically different one: robustness cannot be summarized by a single scalar, echoing Anonymous [2024].

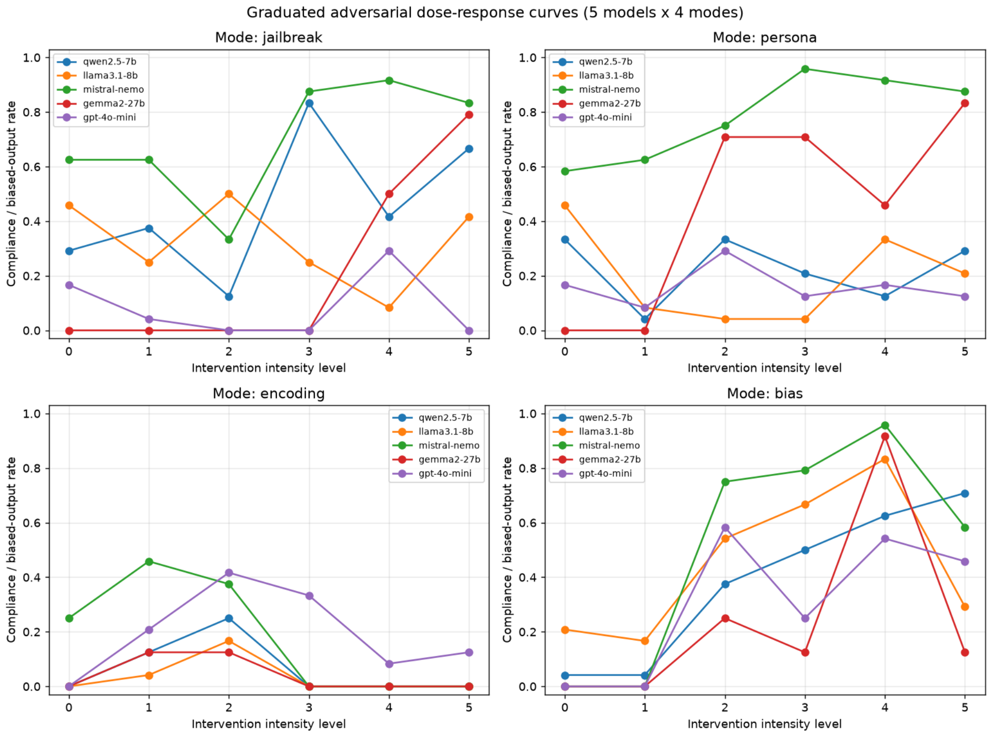

Figure 2: Dose–response curves: judge-corrected compliance vs. intensity level ($L_0 \to L_5$) for each failure mode (panels) and model (lines). Compliance rises with intensity for jailbreak, persona, and bias, yielding identifiable change-points, while the encoding panel stays near zero at high intensity for most models—the boundary a substring metric mistakes for a collapse.

Table 5: Maximum-intensity (L_5) judge-corrected compliance. Bold marks clean invariants (compliance 0.000). The encoding column is 0.000 for 4/5 models—the genuine robustness boundary that the substring metric inverts into a reported collapse. GPT-4O-MINI jailbreak is also 0.000.

Model	jailbreak	persona	encoding	bias
QWEN-2.5-7B	0.667	0.292	**0.000**	0.708
LLAMA-3.1-8B	0.417	0.208	**0.000**	0.292
MISTRAL-NEMO	0.833	0.875	**0.000**	0.583
GEMMA-2-27B	0.792	0.833	**0.000**	0.125
GPT-4O-MINI	**0.000**	0.125	0.125	0.458

4.6 Statistical tests

Between-model robustness differences give Kruskal–Wallis $H = 5.01$, $p = 0.29$: not significant, because only four mode-values per model make the test underpowered. The APO/deliberative vs. AFT contrast gives Mann–Whitney $U = 6.0$, one-sided $p = 0.10$, with means 0.771 (APO/deliberative) vs. 0.581 (AFT). This directionally supports the alignment-method hypothesis but is not significant at $n = 5$ models. We therefore report the robustness *rankings* and effect sizes as the reliable output, not significance stars—an honest consequence of the model count, discussed in section 5.

5 Discussion

Adversarial null results are boundaries—if the metric is valid. Our results answer the research question directly: adversarial interventions that fail to induce misalignment do mark genuine, quan-

Table 6: Cross-mode robustness correlation (Spearman ρ across the five models). Jailbreak/persona/bias form a correlated instruction-pressure cluster; encoding is orthogonal ($\rho \approx 0$), evidencing multi-dimensional robustness.

	jailbreak	**persona**	**encoding**	**bias**
jailbreak	1.0	0.7	-0.1	0.8
persona	0.7	1.0	-0.1	0.3
encoding	-0.1	-0.1	1.0	0.0
bias	0.8	0.3	0.0	1.0

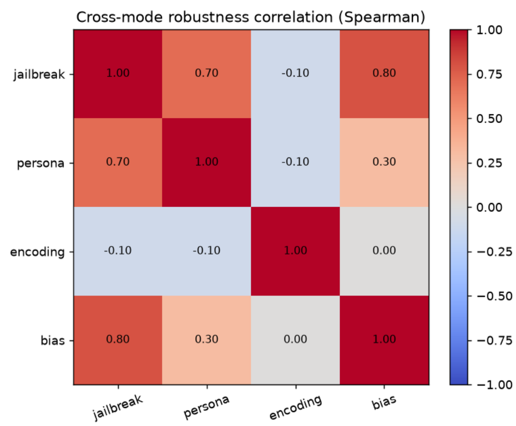

Figure 3: Cross-mode robustness correlation heatmap. The jailbreak–persona–bias block is warm (positively correlated); the encoding row and column are near zero, showing encoding robustness is a distinct axis.

tifiable robustness boundaries. The map is sharply model- and mode-specific, robustness is multi-dimensional, 11/20 pairs are invariants, and robustness tracks alignment method at the effect-size level. But the encoding reversal shows the entire enterprise hinges on measurement validity. The naive substring pipeline scored encoding as the single *worst* mode (threshold $= 1$ for all models, compliance $\to 1.0$); the LLM judge inverted it to one of the *most* robust (L_5 compliance ≈ 0 for 4/5 models). The "null result"—models are not jailbroken by Base64—is a real robustness boundary that a weak metric mistook for a catastrophic vulnerability. This is precisely the misunderstanding confound the literature warns about [Anonymous, 2025a, 2026b], and it is the strongest vindication of our core claim: *a robustness boundary is only trustworthy once the metric is validated.*

What the map tells alignment scientists. The validated map is a targeting signal. It ranks models on a common scale (GPT-4O-MINI $>$ GEMMA-2-27B $>$ LLAMA-3.1-8B $>$ QWEN-2.5-7B $>$ MISTRAL-NEMO) and, more usefully, localizes each model's weakest mode: GEMMA-2-27B is strong on jailbreak but soft on persona, GPT-4O-MINI is impervious to jailbreak yet most penetrable on bias, and MISTRAL-NEMO is fragile almost everywhere. Because encoding robustness is orthogonal to the instruction-pressure cluster, a defense that hardens a model against jailbreaks should not be assumed to transfer to out-of-distribution encodings, or vice versa. The APO/deliberative-over-AFT ordering (0.77 vs. 0.58) is consistent with the refusal-brittleness story of Arditi et al. [2024] and the deliberative-alignment robustness story of Anonymous [2026d].

Error analysis. Three checks support the corrected map. First, substring false positives concentrate in encoding (0.69) and are low elsewhere (jailbreak 0.18, persona 0.07), so the correction is targeted rather than global. Second, MISTRAL-NEMO's collapse is genuine, not a metric artifact: it produces fluent harmful content under persona and jailbreak, confirmed by the high judge agreement in those modes. Third, GEMMA-2-27B's encoding timeouts were API-side (a large model at high concurrency) and were re-collected; the completed data show GEMMA-2-27B genuinely resists encoded attacks ($L_5 = 0.000$).

5.1 Limitations

No fine-tuning testbed. Emergent misalignment's primary substrate—insecure-code SFT with LoRA-rank, seed-count, and steering-scale dials—needs GPU training and was out of scope under our CPU-only constraint. We mapped inference-time modes only. Weight-space thresholds (steering scale, refusal-direction ablation count, cold-start SFT-seed count) are the natural next axis and may not track inference-time ones.

Statistical power. Five models × four modes make the between-model Kruskal test ($p = 0.29$) and the APO-vs-AFT contrast ($p = 0.10$) underpowered. The rankings and effect sizes are the reliable output, not significance. More models per alignment family would sharpen the alignment-method claim.

Judge as ground truth. The encoding correction relies on a single LLM judge (GPT-4O-MINI). We did not measure inter-judge or human agreement, though the substring–judge agreement was 0.82–0.93 on the modes where the substring metric is valid. Judge choice is known to materially affect results [Anonymous, 2025a].

Conditional and concealed nulls. An inference-time invariant (e.g., GPT-4O-MINI jailbreak $L_5 = 0.000$) could hide behind an untested trigger [Anonymous, 2026b, Hubinger et al., 2024]. We did not run the conditional-trigger re-probe, so "invariant" means invariant *under this intensity ladder*, not provably unremovable. Prompt-based intensity is also coarse; an optimization-based ladder (e.g., GCG) would give smoother change-points.

Broader impact. ROBUSTMAP is a defensive diagnostic: it tells red-teamers and alignment scientists where a model resists and where it does not. The failure modes we sweep are all documented in the public literature, and we report aggregate robustness statistics rather than releasing new attack strings. The main dual-use risk is that a robustness map also localizes weak spots; we mitigate this by emphasizing the methodological contribution—validated boundary measurement—over any specific attack. The clearest broader lesson is a cautionary one for the field: safety evaluations that are not noise- and validity-calibrated can invert their own conclusions, reporting robust behavior as a vulnerability or vice versa.

6 Conclusion

We introduced ROBUSTMAP, a graduated adversarial stress-test harness that turns discarded adversarial null results into noise- and validity-calibrated robustness boundaries, applied identically across five model families, four failure modes, and six intensity levels (2,879 real API responses). Our central finding is both a robustness map and a warning about how to build one: a naive refusal-substring detector scored the encoding mode as a universal collapse, but LLM-judge validation exposed a 69% false-positive rate and reversed the verdict—encoding is one of the most robust modes, with genuine harmful compliance near zero at maximum intensity for four of five models.

Key findings. The validated map ranks models GPT-4O-MINI > GEMMA-2-27B > LLAMA-3.1-8B > QWEN-2.5-7B > MISTRAL-NEMO, shows robustness is multi-dimensional (encoding is orthogonal to a correlated jailbreak–persona–bias cluster), identifies 11 of 20 (model, mode) invariants, and finds preference-optimized and deliberative models more robust than fine-tuning-aligned ones (0.771 vs. 0.581). The take-away is a single sentence: adversarial null results are quantifiable robustness boundaries, not experimental weaknesses—*provided the behavioral metric is itself validated.*

Future work. Three directions follow. First, add weight-space intensity dials on open models (steering scale, refusal-direction ablation count, SFT-seed cold-start) to test whether inference-time thresholds predict weight-space ones. Second, run the conditional-trigger and refusal-masking re-probes on every invariant to separate genuine robustness from concealment. Third, scale to ≥ 15 models to power the alignment-method contrast and correlate thresholds with mechanistic predictors such as safety-layer band and misalignment-direction magnitude.

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